Similarly, we can subtract two algebraic expressions. For example, to subtract $13a - 2$ from $25a + 11$, we have to add the additive inverse of $13a-2$ with $25a + 11$.

Additive inverse of $13a-2$ is $-(13a-2) = -13a + 2$

Therefore, $25a + 11 - (13a - 2) = (25a + 11) + (-13a + 2)$

$= (25a - 13a) + (11 + 2)$

$= 12a + 13$

Note

1. Subtracting $4y$ is the same as adding $-4y$ and subtracting $-11x$ is the same as adding $11x$.
2. We know that, -3 is the additive inverse of 3. In the same way $-x$ is an additive inverse of x. Hence, x, $-x$ are equal in numerical value; but opposite in sign. Therefore, $x+(-x) = 0$. But, $x-(-x) = x+x = 2x$.

Example 3.6 Subtract: i) $7pq$ from $11pq$ ii) $-a$ from a iii) $5x + 7$ from $21x + 9$

Solution

i) $11pq - 7pq$. Additive inverse of $7pq$ is $-7pq$

$11pq + (-7pq) = 11pq - 7pq = (11 - 7)pq = 4pq$

ii) $a - (-a)$. Additive inverse of $-a$ is a.

So, $a + a = 2a$

iii) $21x + 9 - (5x + 7)$. Additive inverse of $5x + 7$ is $-(5x + 7)$.

$(21x + 9) + [-(5x + 7)] = (21x + 9) - (5x + 7)$

$= 21x + 9 - 5x - 7$

$= (21 - 5)x + (9 - 7)$

$= 16x + 2.$

Think

$3x + (y - x) = 3x + y - x$. But, $3x - (y - x) \neq 3x - y - x$. Why?.

Example 3.7 Simplify: $100x + 99y - 98z + 10x + 10y + 10z - x - y + z$

Solution In the given algebraic expression, x, y, z are the variables.

Let us group the like terms.

$100x + 99y - 98z + 10x + 10y + 10z - x - y + z$

$= (100x + 10x - x) + (99y + 10y - y) + (-98z + 10z + z)$

$= (100 + 10 - 1)x + (99 + 10 - 1)y + (-98 + 10 + 1)z$

$= (110 - 1)x + (109 - 1)y + (-98 + 11)z$

$= 109x + 108y + (-87)z$

$= 109x + 108y - 87z.$

Note

For adding or subtracting algebraic expressions, we can write the like terms one after the other horizontally by using brackets or one below the other vertically.

7th_Maths_T1_EM_Chp3.indd 60 10/12/2022 18:07:47